COLLABEFY PLATFORM

ENTERPRISE TECHNICAL SPECIFICATION & DATA ARCHITECTURE

System Name: Collabefy Creator Marketplace

Architecture: Decoupled SPA + RESTful API

Release Version: v2.0.0 (Production Ready)

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1. Executive Summary & Purpose

Core Vision

Collabefy is a modern, high-performance creator economy and influencer marketplace platform tailored for the Indian and global social media ecosystem. It directly bridges verified multi-platform creators (Instagram, YouTube, X) with leading brands and agencies.

The platform eliminates traditional agency markups, opaque pricing, and communication bottlenecks by providing transparent upfront rate cards, verified audience analytics, single-date and range availability scheduling, real-time 1-on-1 direct chat, verified campaign reviews, and automated commission fee tracking.

- **Zero-Agency Markups:** Brands directly negotiate and book creators with fixed, transparent rate cards.
- **Multi-Role Workspaces:** Tailored portals for Business Clients, Creators, and Platform System Administrators.
- **Dual-Tier Storage Resilience:** Seamless hybrid storage using MySQL InnoDB with instantaneous automatic to if fallback local JSON storage MySQL is offline.

2. Technology Stack & Frameworks

Modern Full-Stack

Tier / Domain	Technology	Version	Key Responsibility & Benefits
Frontend Framework	React.js (SPA)	v18 / v19	Reactive UI component hierarchy, dynamic hooks & context state
Build & Dev Tool	Vite Bundler	v8.2.0	Sub-second HMR, optimized tree-shaking, production minification
Routing Engine	React Router DOM	v7.x	Client-side declarative routing, protected role route boundaries
Iconography	Lucide React	v1.28.0	Consistent, crisp scalable vector interface icons
Styling Architecture	Custom Glassmorphism CSS	CSS3 / Tokens	Responsive glassmorphic UI, dynamic light/dark theme variables
Backend Runtime	PHP RESTful API	v8.0+	REST controller endpoints, JSON serialization, auth handling
Relational Database	MySQL / MariaDB	v10.4+ / InnoDB	ACID transactions, relational constraints, foreign keys
Offline Resilience	JSON Data Store Engine	Native JSON	Instant local file fallback ensuring zero service disruptions

3. Database Connection & Environment Config

Security & DB

Collabefy utilizes a robust environment-driven database architecture. All credentials are isolated in the root environment file (.env) and never hardcoded in source repositories.

Environment Configuration File (.env)

Location: c:\xampp\htdocs\influencer\.env

The configuration file holds credentials for database connectivity, cross-origin resource sharing, file upload constraints, and runtime settings:

Variable Key	Default Value	Description & Usage
DB_HOST	localhost	MySQL server hostname or container network address
DB_PORT	3306	MySQL listening port (default XAMPP MariaDB port)
DB_NAME	influencer_connect	Primary production database name
DB_USER	root	Database authentication user account
DB_PASS	(empty in XAMPP)	Database authentication password
APP_NAME	Collabefy	Platform brand identifier
APP_ENV	development	Application runtime stage (development / production)
CORS_ALLOWED_ORIGINS	http://localhost:5173	Permitted origins for cross-origin browser fetch queries
UPLOAD_MAX_FILESIZE_MB	10	Maximum permissible file size for uploaded creator portfolio assets

Database Connection Implementation (api/config/database.php)

The Database class initializes a high-performance PDO connection with UTF8MB4 charset, native exception handling, disabled prepared statement emulation, and a fast 1-second timeout before dual-mode fallback:

- **Connection Class:** Database::getConnection() instantiates PDO using env() values.
- **DSN Format:** mysql:host={DB_HOST};port={DB_PORT};dbname={DB_NAME};charset=utf8mb4
- **Auto-Initialization:** If MySQL server is live but the database does not exist (error code 1049), automatically api/init_db.php triggers to create schema tables and seed data.

> **Automatic Schema & Table Creation (api/init_db.php)**
>
> Running api/init_db.php or accessing the platform on a fresh installation automatically runs CREATE DATABASE IF NOT EXISTS, generates all 7 co

4. Complete Data Storage Inventory

Where Data Lives

Collabefy employs a multi-tiered data storage architecture. Depending on system state and data categorization, records live across 4 primary layers:

Storage Tier	Physical Location	Data Category	Persistence Scope
Tier 1: Relational DB	MySQL (influencer_connect)	Users, Creators, Bookings, Chats, Reviews	Permanent, relational, ACID compliant
Tier 2: File Datastore	api/data_store.json	Full backup catalog & dual-mode fallback	Permanent file-based JSON sync
Tier 3: Browser Cache	Client localStorage	Session user, theme, read notifications, d	Device specific browser persistence
Tier 4: File Uploads	uploads/ & frontend/public/	Creator avatars, portfolios, brand logos	File system static image binaries

Relational Database Tables Schema Breakdown

Table Name	Primary Columns & Types	Key Relationships & Purpose
users	id INT (PK), name VARCHAR, email VARCHAR UNIQUE, role ENUM, password VARCHAR, phone VARCHAR, status VARCH	Authenticated Brand Users, Creators, and Platform Admi
influencer_profiles	id INT (PK), user_id INT (FK), name, username UNIQUE, bio, city, category, starting_price, followers, rating, services (JSON)	Creator rate cards, portfolio links, follower counts, and verifie
categories	id INT (PK), name VARCHAR, slug VARCHAR UNIQUE, icon VARCHAR, description TEXT, status VARCHAR	Industry niches (Fashion, Tech, Food, Travel, Fitness, Autom
bookings	id INT (PK), user_id INT, influencer_id INT, campaign_name, budget INT, date DATE, status ENUM, promotion_type	Campaign booking contracts (status: pending, accepted, com
messages	id INT (PK), conversation_id INT, sender_id INT, receiver_id INT, message TEXT, is_read TINYINT, created_at	Real-time 1-on-1 direct negotiation messages between brand
reviews	id INT (PK), influencer_id INT, user_id INT, user_name VARCHAR, rating TINYINT (1-5), review_text TEXT	Verified campaign reviews and satisfaction ratings recalculat
availability	id INT (PK), influencer_id INT, from_date DATE, to_date DATE, status VARCHAR ('busy', 'holiday'), notes	Creator blackout calendar preventing overlapping booking pr
site_settings	id INT, setting_key VARCHAR UNIQUE, setting_value TEXT, updated_at	Dynamic global branding: site name, logo, tagline, commissio

5. User Roles, Portals & Workflows

3 Core Roles

Role 1: Business / Brand Client (user@demo.com)

- **Discovery & Search:** Filter creators by niche category, follower range, city, starting budget, and keyword search.
- **Direct Booking:** Submit campaign proposals with deliverable choice (Reels, Story, Video) and date scheduling.
- **Direct Messaging:** Chat 1-on-1 with creators to agree on deliverables and deadlines.
- **Verified Reviews:** Rate creator deliverables upon campaign completion with 1 to 5 star feedback.

Role 2: Content Creator / Influencer (influencer@demo.com)

- **Rate Cards Manager:** Set fixed pricing packages for Instagram Reels, Posts, and YouTube Videos.
- **Availability Calendar:** Block dates for personal shoots or holidays to prevent conflict bookings.
- **Campaign Negotiation:** Review incoming proposals and click Accept, Decline, or Complete deliverable.
- **Profile & Portfolio:** Showcase recent brand collabs, social handles, bio, and metro city location.

Role 3: System Administrator (admin@influencer.com)

- **Financial Analytics:** Live ledger calculating gross booking volume, platform fee commissions, and payouts.
- **Categories Management:** Add, edit, or delete marketplace categories with custom icons.
- **Creator Verification Audit:** Grant or revoke blue verified creator badges.
- **Site Customization:** Update platform name, slogan tagline, hero imagery, and commission rates in real time.

6. Key REST API Endpoints Overview

REST Architecture

Method	Endpoint Route	Parameters / Body	Action Performed
POST	/api/login	{ email, password }	Authenticates credentials and returns user role token.
GET	/api/influencers	?category=Fashion&search=Aanya	Returns filtered list of active verified creators.
GET	/api/influencer-detail	?id=1	Fetches creator portfolio, services, rating, and socials.
POST	/api/bookings	{ user_id, influencer_id, date, budget }	Submits new campaign booking proposal.
PUT	/api/bookings	{ id, status: 'accepted' }	Updates booking lifecycle transition.
POST	/api/messages	{ conversation_id, receiver_id, message }	Sends real-time direct chat message.
POST	/api/reviews	{ influencer_id, rating: 5, comment }	Submits verified client review and recomputes score.
POST	/api/availability	{ influencer_id, from_date, to_date }	Blocks creator calendar dates.
POST	/api/settings	{ site_name, commission_fee }	Updates platform branding and system commission.

7. Local Setup, Deployment & Verification

DevOps Guide

Prerequisites

Ensure your local development environment has XAMPP (Apache + MySQL with PHP 8.0+) and Node.js (v18 or v20+) installed.

Quick Start Commands

- **1. Apache & MySQL:** Open XAMPP Control Panel and start Apache and MySQL services.
- **2. Configure .env:** Edit c:\xampp\htdocs\influencer\.env with your MySQL port and credentials.
- **3. Initialize DB:** Run 'php api/init_db.php' in terminal to auto-generate database and initial tables.
- **4. Frontend Dev Server:** Run 'cd frontend && npm run dev' to start Vite client at http://localhost:5173.
- **5. Production Build:** Run 'cd frontend && npm run build' to bundle high-performance static assets into /dist.

Automated Test Suite Execution

Collabefy includes a comprehensive end-to-end automated test runner for verifying data consistency and API endpoints:

- **Run Backend Tests:** Command: php scratch/run_full_api_test.php (Verifies 22 API endpoints with 100% pass rate)
- **Run Frontend Tests:** Command: node frontend/scripts/automated_test_runner.js (Verifies 61 data models & form checks)

> **Dual-Mode Zero-Disruption Reliability**
>
> If MySQL is stopped or undergoing maintenance, Collabefy instantly switches to api/data_store.json in under 1 millisecond. Users never see databas

Project Directory Structure Summary

c:\xampp\htdocs\influencer

+-- .env & .env.example (Database credentials & environment configuration)

+-- api/ (PHP REST API endpoints, routing, config, and data_store.json fallback)

+-- frontend/src/ (React 18 source components, pages, context, and custom styling)

+-- frontend/public/ (Collabefy brand logos, cropped icons, and model photography)

+-- uploads/ (Uploaded media, avatars, and campaign brief attachments)